BU MET CS579 - Homework 6

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Problem 1

Write the following queries in relational algebra, which are issued to the Company database.

(1) Show the first name and last name of every employee who work for the Research department:

$$Relation_1 \leftarrow employee \bowtie_{dno=dnumber} department$$

$$Relation_2 \leftarrow \sigma_{dname='Research'}(Relation_1)$$

$$Result \leftarrow \Pi_{fname,lname}(Relation_2)$$

(2) Show the first name and last name of every employee who works on a project that is located in Houston:

$$Relation_1 \leftarrow \sigma_{plocation='Houston'}(project)$$

$$Relation_2 \leftarrow \sigma_{ssn=essn}(employee \times (\sigma_{pno=pnumber}(works_on \times Relation_1)))$$

$$Result \leftarrow \Pi_{fname,lname}(Relation_2)$$

(3) Retrieve the first name and the last name of every employee who works for the Research department and works on ProductZ project:

$$Relation_1 \leftarrow \Pi_{dnumber}(\sigma_{dname='Research'}(department))$$

$$Relation_2 \leftarrow \Pi_{essn}(\sigma_{pname='ProductZ'}(\sigma_{pno=pnumber}(works_on \times project)))$$

$$Relation_3 \leftarrow \sigma_{dnumber=dno}(Relation_1 \times (\sigma_{ssn=essn}(employee \times Relation_2)))$$

$$Result \leftarrow \Pi_{fname,lname}(Relation_3)$$